

Understanding the acceptance of business intelligence from healthcare professionals' perspective: an empirical study of healthcare organizations

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Abstract

Purpose – Due to its ability to support well-informed decision-making, business intelligence (BI) has grown in popularity among executives across a range of industries. However, given the volume of data collected in health-care organizations, there is a lack of exploration concerning its implementation. Consequently, this research paper aims to investigate the key factors affecting the acceptance and use of BI in healthcare organizations.

Design/methodology/approach – Leveraging the theoretical lens of the “unified theory of acceptance and use of technology” (UTAUT), a study framework was proposed and integrated with three context-related factors, including “rational decision-making culture” (RDC), “perceived threat to professional autonomy” (PTA) and “medical-legal risk” (MLR). The variables in the study framework were categorized as follows: information systems (IS) perspective; organizational perspective; and user perspective. In Jordan, 434 healthcare professionals participated in a cross-sectional online survey that was used to collect data.

Findings – The findings of the “structural equation modeling” revealed that professionals' behavioral intentions toward using BI systems were significantly affected by performance expectancy, social influence, facilitating conditions, MLR, RDC and PTA. Also, an insignificant effect of PTA on PE was found based on the results of statistical analysis. These variables explained 68% of the variance (R^2) in the individuals' intentions to use BI-based health-care systems.

Practical implications – To promote the acceptance and use of BI technology in health-care settings, developers, designers, service providers and decision-makers will find this study to have a number of practical implications. Additionally, it will support the development of effective strategies and BI-based health-care systems based on these study results, attracting the interest of many users.

Originality/value – To the best of the author's knowledge, this is one of the first studies that integrates the UTAUT model with three contextual factors (RDC, PTA and MLR) in addition to examining the suggested framework in a developing nation (Jordan). This study is one of the few in which the users' acceptance behavior of BI systems was investigated in a health-care setting. More specifically, to the best of the author's knowledge, this is the first study that reveals the critical antecedents of individuals' intention to accept BI for health-care purposes in the Jordanian context.

Keywords Business intelligence, Health care, UTAUT, Developing nations, SME

Paper type Research paper



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1. Introduction

Health care today offers a growing market for cutting-edge technology. The adoption of novel devices, practices, data analytics techniques and other advancements have empowered advanced IT/IS, such as the “Internet of Things (IoT) and Artificial Intelligence (AI),” to facilitate the delivery of improved services that are critical to individuals’ lives ([Ben Arfi et al., 2021](#); [Gadeikienė et al., 2021](#); [Asad et al., 2022](#); [Alshaheer et al., 2022](#); [Salhab et al., 2023](#); [Alharasis et al., 2023](#); [Shehadeh et al., 2023](#)). Using “information technology” (IT) in the health-care industry effectively could result in cost reduction and organizational restructuring, which could assist citizens in receiving cost-effective health-care services ([Ain et al., 2019](#)). Business intelligence (BI) systems are normally known as a suite of IT solutions that assists organizations in amassing, integrating and analyzing massive stocks of data to understand their external opportunities and threats, besides their internal strengths and weaknesses ([Salisu et al., 2021](#)). In practice, BI combines “business analytics, data mining, data visualization, data tools and infrastructure, and best practices to help organizations make more data-driven decisions” ([Tableau, 2023](#)). In the 1990s, Howard Dressner, an Analyst at Gartner Group, first coined the concept of “business intelligence” ([Ain et al., 2019](#); [Bhatiasevi and Naglis, 2020](#); [Salisu et al., 2021](#)). Researcher and practitioner debates regarding the strategic and tactical procedures toward efficient acceptance and use of BI systems have been shown to be the driving force behind recent advances in BI research initiatives. However, a limited number of published articles have aimed to investigate these systems in a specific sector, for example, health care. In the health-care industry, BI can be described as the usage of information and dedicated analytical tools to support an informed decision-making process ([Foshay and Kuziemyky, 2014](#)).

The health-care sector is under ever-widening pressure to make greater efforts and is constantly pursuing strategies to guarantee the most appropriate distribution and allocation of resources while guaranteeing high-standard care for patients ([Foshay and Kuziemyky, 2014](#); [Churi et al., 2022](#)). To achieve this, IT/IS is essential to transform data into intelligence that can be used to enhance process management, health-care infrastructure and patient care ([Gastaldi et al., 2018](#)). A crucial feature of BI systems is the ability to integrate data from a variety of interior and exterior sources, hence offering decision-makers in the health-care industry valuable information platforms ([Cunha et al., 2023](#); [Ramakrishnan et al., 2020](#)). It is widely accepted that BI could assist health-care institutions by improving care and outcomes for their patients, optimizing the practice of “human resources” (HR), increasing the efficiency of processes and reducing costs ([Foshay and Kuziemyky, 2014](#); [Kitsios and Kapetaneas, 2022](#)). BI has emerged as a key field of study for both scholars and practitioners due to the vital role that data plays in driving the development of health-care organizations. In reality, BI could not only enhance the performance of health-care institutions but also benefit them in advancing along the spectrum from intuitive to precision medicine “personalized medicine” ([Basile et al., 2023](#); [Gastaldi et al., 2018](#)).

BI is growing globally at a breakneck speed. These technology advancements such as “business intelligence” have become vital for organizations to improve their management techniques, outcomes, products and services as a result of growing competition from offline and online industries ([Ain et al., 2019](#); [Hmoud et al., 2023](#)). Due to its ability to enhance the decision-making process with more intellectual information, BI systems have recently gained a lot of interest from executives and decision-makers ([Bhatiasevi and Naglis, 2020](#); [Salisu et al., 2021](#)). According to [Gartner \(2017\)](#), the market for BI is anticipated to reach a value of \$18.3 billion in 2017 and approach \$22.8 billion by 2020 ([Gartner, 2017](#)). Additionally, in 2016, BI revenue worldwide amounted to \$17.09 billion and is projected to grow by 9.5% over the next five years to reach \$26.88 billion by 2021 ([Gottfried et al., 2021](#)). More organizations are transforming to data for decision-making, which is driving up the need for analytics and BI technologies. Also, platforms for BI and data analytics are

developing swiftly to keep up with the demands of a growing user base that comprises both technical and non-technical end-users. Analysis and BI software market size is expected to double by 2025, expanding at a 16% compound annual growth rate from 2020 to reach \$13bn, predicts Gartner research (Gartner, 2022). Health care is one of the areas where BI has the most promise because it is one of the key industries where technology might have a noteworthy effect on the decision-making process and outcomes.

The COVID-19 crisis has caused substantial shifts in many industries worldwide. In addition to the many profound shifts that have occurred in society and in human affairs, there has also been a great deal of new opportunities in the business, educational and health-care sectors (Čartolovni *et al.*, 2022; Gadeikienė *et al.*, 2021; Rahi *et al.*, 2021). Like other technological solutions, the adoption of BI in the health-care industry is predicted to undergo a real rebirth post the pandemic era, as evidenced by the growing number of studies and applications in the domain (Basile *et al.*, 2021; Sechi *et al.*, 2020). In this industry, BI is perceived as a major advancement to enhance conventional decisions made by doctors or health-care practitioners (Gastaldi *et al.*, 2018). However, despite the fact that there are various applications based on the usage of data to enhance health-care procedures (such as assisting doctors in choosing and monitoring prognosis and diagnosis) (Basile *et al.*, 2023; Foshay and Kuziemy, 2014), or the IT/IS architectures, the “management information systems” (MIS) and “database management systems” (DBMS) in the health-care industry (Bez *et al.*, 2023; Janyapoon *et al.*, 2021), the use of IT solutions (e.g. BI) is still limited, thus it appears that there is still room for improvement in the use of data for health-care management operations (Basile *et al.*, 2023; Olszak and Batko, 2012; Salisu *et al.*, 2021).

In recent years, health-care organizations have presented an increasing tendency toward the acceptance and use of BI technologies (Basile *et al.*, 2023; Kitsios and Kapetaneas, 2022). This inclination could be explained by the continuous developments in tools and techniques of data analytics that enable health-care organizations to manage patients’ data and other health-care-related information efficiently. According to Gastaldi *et al.* (2018), health-care organizations that can go through transformational changes thanks to technological innovation, like BI, will witness advances in all aspects of their infrastructure, health-care management competency and ability to meet future needs. This assists the health-care sector in thoroughly examining recently emerging difficulties, exploring and identifying innovative solutions to these challenges and predicting potential outcomes to encourage growth in the future (Abed *et al.*, 2022).

However, despite the increasing investment, market expansion and significant benefits of BI adoption, the evidence has pointed out that BI systems still experience substantial failure rates (Hmoud *et al.*, 2023; Jaradat *et al.*, 2022). More than 70% of BI initiatives fail to deliver the expected outcomes or provide organizations with minimal or no benefits (Salisu *et al.*, 2021). Earlier literature pointed out that the failure to implement BI systems could be ascribed to the lack of understanding of the factors affecting BI acceptance and use (Ain *et al.*, 2019; Kašparová, 2023; Rouhani *et al.*, 2018). Although there is growing research interest in BI acceptance and use in the health-care arena (Janyapoon *et al.*, 2021; Kitsios and Kapetaneas, 2022), there have not been extensive studies conducted in this context, especially in developing nations. Thus, the factors affecting the behavioral intention toward using BI for health care from patients’ perspectives is still a research phenomenon that has received little attention in the Middle East region in general and Jordan is no exception. The aim of this research is to investigate the intention to adopt BI for health-care purposes using evidence from the Jordanian context. To achieve this aim, the current study develops a study model grounded on the “unified theory of acceptance and use of technology” (UTAUT), which is well-known for its ability to explain the aspects that influence the acceptance and use of innovative technologies (e.g. BI) at the users’ level to tackle the subsequent research

question: What variables affect the users' intentions toward BI systems in Jordanian health-care organizations?

To this end, this research paper contributes to the existing BI research by determining the variables that drive the acceptance and use of BI systems in the health-care setting. Unlike other areas, the health-care field has demonstrated a slower pace in adopting this innovative technology, which raises concerns about the willingness of users in health-care organizations to accept and use BI, particularly in developing countries. Furthermore, this study aims to offer executives and decision-makers in the health-care sector valuable guidance and directions that could facilitate the effective and successful application of BI systems and ensure efficient outcomes.

The remainder of this paper is organized as follows: Section 2 provides a review of the BI in health care. Section 3 presents the theoretical model and hypotheses. Section 4 illustrates the research methodology applied in this research. Section 5 presents the findings of the statistical analysis. Sections 6 and 7 discuss the findings and the implications for theory and practice. Finally, additional promising research directions are delineated and limitations are offered.

2. Business intelligence in health care

Health-care organizations are constantly under pressure to improve results while using fewer resources (Basile *et al.*, 2023) and to transform into information-driven systems over time (Gastaldi *et al.*, 2018). Because of the swift growth in the amount of data being collected by medical registries and "electronic health records" (EHRs), health-care organizations are attempting to leverage tools like BI systems to boost their effectiveness as well as their productivity (Arefin *et al.*, 2021; Basile *et al.*, 2023; Gastaldi *et al.*, 2018; Kitsios and Kapetaneas, 2022). Although BI was first considered merely as a set of data analysis tools, over time, many organizations have grown to perceive it as a strategic weapon (Gastaldi *et al.*, 2018). Today, BI is usually considered an umbrella term for a set of theories, approaches, procedures and methods for transforming data into valuable and insightful information for business uses (Kašparová, 2023). BI systems support decision-makers, by offering valuable insights in the right format, at the right place and at the right time (Bernardino *et al.*, 2019; Nyanga *et al.*, 2020).

As recently reported (Gartner 2017, 2022), the market for BI has expanded dramatically and has risen to the top of the "Chief Information Officers'" list of priorities for investment. The general awareness of its potential positive effects is also growing (Jaradat *et al.*, 2022). Yet, the adoption of BI in the health-care industry moves slowly and on an as-needed basis (Basile *et al.*, 2023; Foshay and Kuziemy, 2014). Limited research efforts conducted to address this research phenomenon. In their research, Foshay and Kuziemy (2014) sought to provide answers to the following questions: "(1) what are the most significant adverse impacts to the organization's decision processes and outcomes attributable to a lack of decision-support capabilities? (2) what are the root causes of these impacts, and what workarounds do they necessitate? and (3) In light of the issues identified, what are the key considerations for health-care organizations in the early stages of BI implementation?" In contrast, the aim of Basile *et al.*'s (2023) study was to ascertain if a decision support system (DSS) model that leverages data through BI can manage operations in the health-care industry more effectively than traditional experience-driven methods. Most of the earlier literature was focused only on technological or medical dimensions, while the factors affecting the acceptance and use of BI for enhancing health-care practices are still under-investigated arena.

Understanding and examining the unique characteristics of this industry is critical for the effective implementation of BI in health care. Consequently, the key aim of this research is to provide health-care professionals with a study model for developing and continuously enhancing their BI solutions based on the variables affecting the adoption of this ground-breaking IT.

2.1 Health-care system and business intelligence in Jordan

Jordan has been actively developing its electronic health-care system to improve health-care delivery and patient outcomes. The country has been investing in various e-health initiatives to enhance efficiency, accessibility and the quality of health-care services (Jalghoum *et al.*, 2021). One of the notable initiatives is the Jordan Health Information System (JHIS), which aims to digitize patient records, streamline administrative processes and facilitate information exchange among health-care providers (Jabareen *et al.*, 2020). JHIS includes EHRs that store patients' medical history, diagnoses, medications and treatment plans, allowing authorized health-care professionals to access and update this information.

In 2009, Electronic Health Solutions (EHS) initiated its flagship program, Hakeem, under the patronage of His Majesty King Abdullah II Bin Al Hussein (Jalghoum *et al.*, 2021; Othman and Hayajneh, 2015). Hakeem is the first national digital transformation initiative in Jordan's health-care sector. The objective of the Hakeem initiative is to enhance the delivery of effective and superior health-care services in the Kingdom by implementing an EHR system across the entire country (EHS, 2021). Using electronic systems designed in accordance with the most advanced international medical standards in the health-care industry (EHS, 2021; Othman and Hayajneh, 2015; Tubaishat, 2018). Physicians, pharmacists, medical technologists and other clinicians have the ability to electronically retrieve medical records of patients at participating health-care facilities across the county by inputting the patient's national ID number.

Telemedicine and telehealth services have also been expanding in Jordan, especially in remote areas where access to health-care services might be limited (Al-Samarraie *et al.*, 2020). These services enable patients to consult health-care providers remotely through digital platforms, improving access to medical advice and reducing the need for physical visits in some cases. Interoperability and data security are crucial aspects of Jordan's electronic health-care system to ensure that different health-care facilities and systems can communicate and share patient information securely (Jalghoum *et al.*, 2021).

In the context of using "business intelligence" technologies, the "Health Data Analytics" (HDA) Program was officially introduced in Jordan under the patronage of His Majesty King Abdullah II in 2019 to facilitate the extraction of trustworthy information in an integrated and thorough manner by providing technical and artificial solutions, data mining tools and putting analytical solutions into practice for better decisions. HDA was developed by the "Electronic Health Solutions" (EHS) team to assist and empower academics, researchers, practitioners and decision-makers in the health-care field. To achieve its principal aim of supporting the Jordanian health sector, the "HDA" was built around three significant pillars: boosting research and studies, enhancing data quality and providing "business intelligence" solutions (EHS, 2020; Petra, 2021).

Countries with health-care systems similar to Jordan's and an interest in electronic health-care systems might include those within the Middle East region, such as the United Arab Emirates, Qatar and Saudi Arabia, which have been investing in technology-driven health-care solutions (Katoue *et al.*, 2022). Additionally, countries aiming to enhance their health-care infrastructure and adopt digital health initiatives in regions like Southeast Asia

or parts of Africa might also find insights from Jordan's experiences in developing its electronic health-care system.

3. Proposed theoretical model and development of hypotheses

3.1 *Extended unified theory of acceptance and use of technology model for business intelligence system in health-care organizations*

Venkatesh *et al.* (2003) carried out an empirical comparison of eight competing theories to understand how people adopt new technologies: the "theory of reasoned action" (TRA), "technology acceptance model" (TAM), "theory of planned behaviour" (TPB), combined TAM and TPB, "innovation diffusion theory" (IDT), "motivational model" (MM), "model of PC utilization" (MPCU) and "social cognitive theory" (SCT). 215 participants from four organizations were surveyed. Venkatesh *et al.* (2003) combined and expanded on these eight models based on their longitudinal investigations to develop a new theory known as the UTAUT. This study uses the UTAUT model as a theoretical foundation for the formulation of the proposed hypotheses because it has been empirically validated and demonstrated to be superior to other theoretical frameworks (Venkatesh *et al.*, 2003; Venkatesh and Zhang, 2010).

The UTAUT model comprises four significant dimensions that affect behavioral intentions to use an IT/IS and/or IT/IS use, namely, "performance expectancy, effort expectancy, social influence, and facilitating conditions." These concepts and terminology from the UTAUT are adjusted by the researchers for the adoption of various technologies (Dwivedi *et al.*, 2019; Venkatesh *et al.*, 2016; Alkhwaldi and Al Eshoush, 2022). The UTAUT model was initially established to investigate how technologies are adopted in the workplace. It was then expanded to examine the variables that affect how innovations are adopted at the individual level in various settings (Venkatesh *et al.*, 2012a, 2012b). For example, the UTAUT has been applied to determine the variables influencing the acceptance and use of m-health from a developing country perspective (Barua and Barua, 2021), e-health care from the viewpoint of the IoT natives and immigrants (Ben Arfi *et al.*, 2021) and clinical DSS (Zha *et al.*, 2022). As a result, the UTAUT is typically expanded by additional contextual factors that include particular aspects from the usage area (Venkatesh *et al.*, 2012a, 2012b), such as the "Anthropomorphism" to predict the intention to use Chabot (Melian-González *et al.*, 2021), or privacy, trust and personal innovativeness in the area of e-Health (Gu *et al.*, 2021). The incorporation of more contextual predictors allows for an improved understanding of users' adoption of technological advancements, which is always domain-specific (Lian, 2015; Venkatesh *et al.*, 2012a, 2012b; Alkhwaldi and Al-Ajaleen, 2022). Since its development, the UTAUT has been widely used as a theoretical focus by academics examining empirical research on users' intentions and behavior in the area of IT/IS literature on acceptance and adoption. Dwivedi *et al.* (2019) state that the UTAUT has successfully harmonized the research on how people adopt new technologies. Furthermore, recently published research reported that the complexity of decision-making processes, BI tools and artificial intelligence-based systems within the health-care context is derived from different sources, such as patients and their medical conditions, health-care professionals and how they interact with other professionals and work procedures within and between settings (Fan *et al.*, 2020; Kašparová, 2023).

The UTAUT appears to be pertinent in this study due to its validity in earlier research as the factors of the original theory have been investigated and validated in the health-care systems (Barua and Barua, 2021) and BI (Kašparová, 2023). Additionally, several researchers have revealed limited application of the UTAUT in areas like medicine and psychology. So, they recommend extending the UTAUT to different and additional areas of research that are ignored and need more attention (Dwivedi *et al.*, 2019; Williams *et al.*, 2015).

In the same regard, the adoption rate of UTAUT in BI research was relatively low. According to a “systematic literature review” (SLR) carried out by [Ain *et al.* \(2019\)](#), only three studies in the past 20 years have shown the original UTAUT model’s contribution to BI adoption research. However, examination of the UTAUT model used across many studies indicates that the most typical limitation is concentrating on a specific concern, for instance, a culture or an organization. The aim of this research is to bridge these gaps.

In the current study, the individual perspective and the BI systems in the health-care setting must be taken into consideration while adjusting the measurement scales because the UTAUT was originally developed to evaluate the adoption of IT in an organizational setting. In addition, as this research is specific to BI in health care, extended specific factors need to be taken into account in the study framework. For example, “medico-legal risk,” (MLR) “rational decision-making culture” (RDC) and “perceived threat to professional autonomy” (PTA) constructs may affect users’ willingness to adopt BI for decision-making in a health-care context.

Drawing from [Ain *et al.*’s \(2019\)](#) SLR, the selected research articles were analyzed further to identify the theoretical variables that the adoption of BI technology. Considering the categorization technique by [Hwang *et al.* \(2004\)](#), the analysis shows three key constructs categories, as presented in [Figure 1](#). First, is the “organizational perspective,” which concentrates on how BI systems align with business strategy, objectives and priorities besides its impact on BI acceptance and use in different organizations. In this study, this category takes account of organization-related constructs, including “social influence,” “facilitating conditions,” “medico-legal risk” and “rational decision-making process.” Second, the “information system (IS) perspective” concentrates on the significance of having a flexible and adaptable IT infrastructure, a system interface that is easy to use (user-friendly), as well as superior data and source systems for BI systems’ acceptance and use. In this study, the “IS perspective” shows the effect of IS-related constructs, including “performance expectancy” and “effort expectancy.” Third, the “users’ perspective” shows the investigation of individual-related constructs. In their SLR, [Ain *et al.* \(2019\)](#) stated that the analysis of these main factors in the BI context found that individual-related constructs have been basically neglected in the literature of BI systems’ acceptance and use. Few published research papers have considered users’ perspectives. The focus of the majority of reviewed studies has remained limited to either “organizational or IS-related factors” as the most important factors influencing BI acceptance and use. In this study, “perceived threat to professional autonomy” was incorporated into the proposed research model as one of the main “users’ perspective” constructs relevant to the adoption of BI in the health-care setting:

- PE

From the UTAUT perspective, “performance expectancy” (PE) is described “as the degree to which an individual believes that applying the technology will help him or her to attain gains in job performance” ([Venkatesh *et al.*, 2003](#)). In the literature of health-care technologies, medical professionals will be driven toward accepting and adopting novel technologies if they think it would be more beneficial and efficient for their clinical practice and health care in general than what is currently being used ([Alam *et al.*, 2020](#); [Barua and Barua, 2021](#); [Alkhwaldi and Abdulmuhsin, 2022](#); [Alkhwaldi *et al.*, 2023](#); [Abu-AlSondos *et al.*, 2023](#)). According to [Esmaeilzadeha *et al.* \(2015\)](#), PE exerts a significant impact on physicians’ intentions to adopt “clinical decision support systems” (CDSS). Also, [Kašparová \(2023\)](#) PE encourages users to adopt BI systems. Therefore, this study argued that practitioners are inclined to develop intentions to use BI health-care systems if they believe that doing so will help them perform their jobs more effectively (i.e. increase their speed,

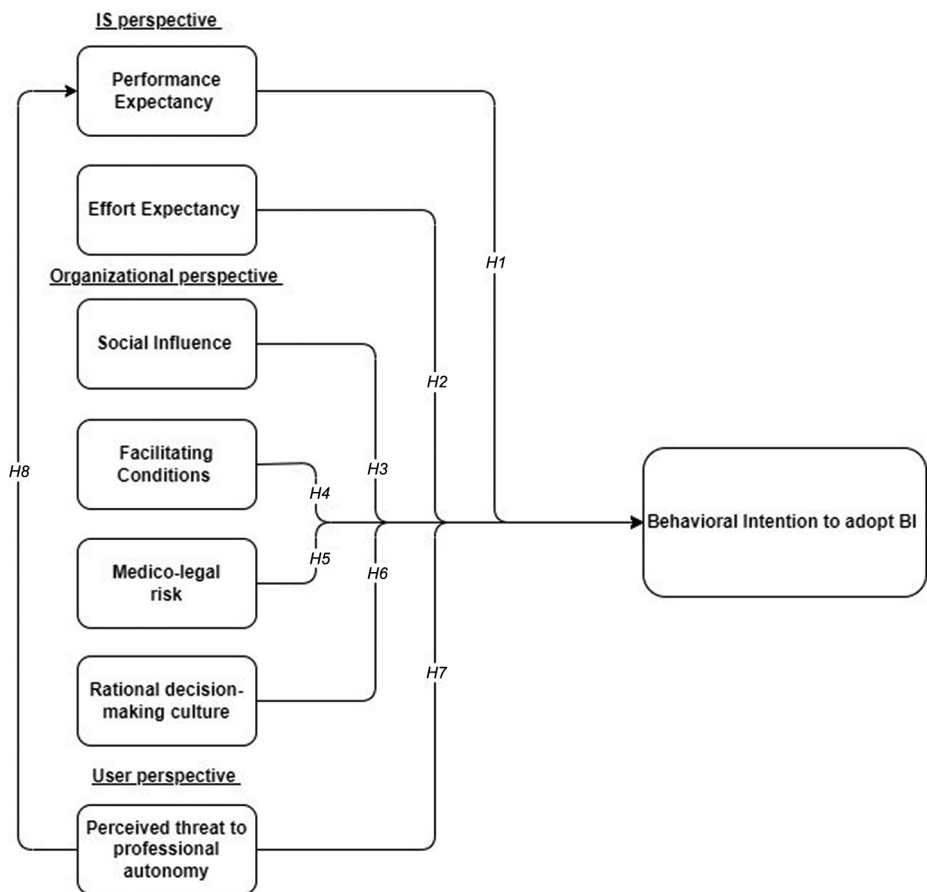


Figure 1.
Study model

Source: Created by author

accuracy and productivity, and minimize their burden). Therefore, the subsequent relationship was hypothesized:

H1. PE will have a positive effect on the behavioral intention to use BI systems in the health-care context.

- EE

According to the UTAUT model, “effort expectancy” (EE) refers to “the extent of ease associated with the use of a system” (Venkatesh *et al.*, 2003). EE was acknowledged in earlier research as a construct enabling the adoption of different health-care technologies (Alam *et al.*, 2020; Prakash and Das, 2021; Dbesan *et al.*, 2023). Users believe that simple “easy to use” health-care systems and applications would be more likely to be used than the complicated and intricate ones (Wang *et al.*, 2020). Also, Gu *et al.* (2021) stated that e-health-

care systems having ease features benefit individuals in developing nations and generate positive changes in their behavioral intention to adopt such systems (Gu *et al.*, 2021). Consequently, in the context of BI health-care systems, it is expected that if the medical practitioners identify the BI systems to be effortless and easy to run, they will be more inclined to develop positive intentions to use it in their health-care practices and health-care decision-making. Thus, the subsequent hypothesis was formulated:

H2. EE will have a positive effect on the behavioral intention to use BI systems in the health-care context.

- SI

“Social influence” (SI) refers to “the degree to which an individual perceives that important others believe he or she should use the new system” (Venkatesh *et al.*, 2003). The UTAUT perspective advocates that the users’ perceptions of the belief of important others (e.g. peers, colleagues, supervisors and friends) concerning whether they need to use a target IT/IS affects their intentions to do so (Venkatesh *et al.*, 2012a, 2012b; Alkhwalidi *et al.*, 2023). Additionally, it has been noted that when adopting novel and innovative practice tools, medical professionals often seek their professional peer groups for approval and endorsement (Gu *et al.*, 2021; Prakash and Das, 2021; Rahi *et al.*, 2021). Recent research by Zha *et al.* (2022) has highlighted the importance of SI in encouraging nurses to use a “clinical decision support system (CDSS)” (Zha *et al.*, 2022). Therefore, this study suggests that the medical Practitioners’ intentions to use BI systems could be affected by the beliefs of others in their social and professional surroundings. The impact of SI could be even more evident in the context of collectivist societies, such as Jordan (Alalwan *et al.*, 2018). Hence, the subsequent hypothesis was formulated:

H3. SI will have a positive effect on the behavioral intention to use BI systems in the health-care context.

- FC

“Facilitating conditions” (FC) describes “the degree to which an individual believes that an organizational and technical infrastructure exists to support the use of the system” (Venkatesh *et al.*, 2003). Thus, FC in health-care technologies is elucidated as the extent to which individuals think that vendors of such technologies will provide IT infrastructure and technical support to adopt e-Health care various systems (e.g. BI systems). Users’ IT skills for using software applications are developed when there is an adequate technological infrastructure available (Moudud-Ui-Huq *et al.*, 2021; Alkhwalidi, 2023), and highly skilled users have demonstrated that they are more inclined to accept and use information technologies (Venkatesh *et al.*, 2012a, 2012b). The availability of IT infrastructure is thus predicted to provide users with technical skills and knowledge on how to use BI systems in the decision-making process within health-care organizations. Hossain *et al.* (2019) found that the availability of IT resources, the required skills and system compatibility have a significant effect on users’ adoption of health-care technologies (Hossain *et al.*, 2019). The present study therefore suggests that FC supports individuals’ behavioral intentions toward BI applications in the health-care industry. Thus, the subsequent relationship was hypothesized:

H4. FC will have a positive effect on the behavioral intention to use BI systems in the health-care context.

- MLR

Another key uncertainty involved in the usage of BI systems in health-care practices is linked to various medical practices such as the “medico-legal” implications of AI-assisted diagnosis. Respondent’s answers to the open-ended question included in this study’s questionnaire provided a corroborative perspective, for example, “Who is going to bear complete liability event of a BI error or incorrect interpretation? People still want a radiologist to overread or file a lawsuit.” The introduction of BI in health care also poses difficult legal issues about the responsibility of health-care practitioners versus manufacturers, particularly if they are unable to explain the system’s recommendations. This is because the nature of BI algorithms which described to be opaque ([Čartolovni et al., 2022](#)). Also, current laws and regulations in most developing countries are ambiguous regarding who is responsible for damages in the case of an injury or damage, which makes the issue worse. Therefore, the medical professional signing the report will ultimately be responsible for any liability resulting from a faulty interpretation or misdiagnosis, not the application developer, designer or programmer who created the BI system ([Chung and Zink, 2017](#); [Mettler and Vimarlund, 2009](#); [Schweikart, 2020](#)). Doctors and other medical practitioners might not want to consent to a new style of practicing using the BI systems as the current legal frameworks are unable to manage the realities of these innovative technologies ([Chung and Zink, 2017](#)). This would increase their risk of being sued for medical malpractice. As a result, practitioners who believe they face a high level of “medico-legal risk” will rather maintain the status quo than adopt the new BI-based practice model in the health-care setting. Thus, the following association was hypothesized:

H5. “Medico-legal” risk will have a negative effect on the behavioral intention to use BI systems in the health-care context.

- RDC

The acceptance of quantitative data measurement, testing and evaluation in decision-making processes across the entire business is a sign of a sane decision-making culture. Such a culture encourages the use of data and information to facilitate operational processes and conduct analysis while also using cutting-edge technologies ([Hoang and Bui, 2023](#)). According to earlier research, organizational culture has a significant beneficial impact on the manner in which an IT innovation is adopted ([Hmoud et al., 2023](#); [Horani et al., 2023](#)). [Popović et al. \(2012\)](#) noted the significant effect of fact-based decision-making culture on the usage of BI systems while studying the impact of organizational culture on the acceptance of innovative technologies ([Popović et al., 2012](#)). According to earlier research, the adoption of big data analytics is positively correlated with the practice of rational decision-making ([Puklavec et al., 2018](#); [Youssef et al., 2022](#)). Consequently, to realize the potential advantages of BI systems and realize the intended results, health-care organizations in Jordan need to cultivate a solid decision-making culture. This may entail fostering a culture of information-driven decision-making, stimulating the dissemination of knowledge and best practices throughout the institution and offering BI users training and assistance. This research proposes that Jordanian health-care organizations should nurture a RDC to encourage the acceptance of BI. Therefore, the subsequent hypotheses were posited:

H6. RDC will have a positive effect on the behavioral intention to use BI systems in the health-care context.

- PTA

PTA can be described as “professionals having control over the conditions, processes, procedures or content of their work according to their own collective and, ultimately, individual judgment in the application of their profession’s body of knowledge and expertise” (Walter and Lopez, 2008), p. 208). BI-based systems could help in the decision-making process in the health-care context. For example, by providing diagnostic and cure alternatives that could be produced based on patient data and records using some investigative reasoning methods and heuristic rules (Esmailzadeh *et al.*, 2015; Walter and Lopez, 2008). Medical doctors and practitioners in health-care organizations may perceive BI systems as an imminent threat to their professional autonomy as offering prescription advice for care planning and giving recommendations during counselling sessions with patients can be perceived as intrusions on their professional autonomy (Harper *et al.*, 2023; Sambasivan *et al.*, 2012). The key barrier to the successful adoption of AI/BI-assisted systems is their acceptance and use by users (e.g. health-care professionals). While professionals’ resistance to adopting AI/BI-assisted systems is somewhat associated with the PTA (Khanijahani *et al.*, 2022).

In accordance with previous research, medical practitioners and doctors will categorically reject any IT/IS that threatens their professional autonomy (Alohali *et al.*, 2020; Harper *et al.*, 2023; Khanijahani *et al.*, 2022; Sambasivan *et al.*, 2012). The successful implementation of AI/BI-based systems in the health-care setting depends on reducing the perception of threat to doctors’ professional autonomy (Eltawil *et al.*, 2023; Moxey *et al.*, 2010). Thus, if medical practitioners and doctors consider the adoption of BI in health-care establishments as a threat to their “professional autonomy” then their intentions to adopt BI decline significantly. One of the strongest predictors of individuals’ intentions to use novel technologies that are immersed in the UTAUT is “performance expectancy (PE)” which is the extent to which medical doctors and practitioners think that using the BI systems would support them realize improvements in their job performance (Venkatesh *et al.*, 2003). PE is significant for health-care/medical professionals to the extent that BI systems do not invalidate their “professional autonomy” (Esmailzadeh *et al.*, 2015). This research hypothesized that PTA is directly associated with intentions to use BI systems. Furthermore, this factor was hypothesized to have a negative association with PE. According to the abovementioned discussion, the current research developed the subsequent hypotheses:

- H7. “Perceived threat to professional autonomy” will have a negative effect on the behavioral intention to use BI systems in the health-care context.
- H8. “Perceived threat to professional autonomy” will have a negative effect on the “performance expectancy” related to using BI systems in the health-care context.

In addition, intentions to use a certain IT does not necessarily translate into the actual use of that technology. The research across the disciplines of sociology, technology and psychology claims that the “behavioural intention” of individuals to refrain from performing or perform an identified behavior could be considered the best predictor of carrying out a certain action (Ajzen and Fishbein, 1980; Venkatesh *et al.*, 2003). This presupposition has been widely confirmed in IT/IS research (Alalwan *et al.*, 2017; Taylor and Todd, 1995). Drawing on the consistent findings of the earlier studies that “behavioural intention” is a determinant of the actual behavior of people, the literature has examined factors that could influence “behavioural intention” to use rather than actual usage. Hence, the current study investigates the respondents’ “behavioural intention” to accept BI-based health-care systems based on a modified UTAUT model.

“Behavioral intention” is defined as the commitment level that the user shows to become involved in a particular behavior (Ngai *et al.*, 2007), which in the context of this study is the BI users’ commitment level to accept the use of a system for health-care purposes. Based on the prior discussion, the research model was conceptualized (see Figure 1). The reason for dropping the demographic characteristics from the proposed model was primarily to present a parsimonious model that could be tested for any situation of using BI for health care, rather than depending too much on any specific context (such as age, gender and other demographics), as models like the UTAUT (Venkatesh *et al.*, 2003) and the UTAUT2 (Venkatesh *et al.*, 2012a, 2012b) do.

4. Methodology

4.1 Sample and data collection

The current study applied a positivist quantitative approach, and data were collected using an online survey via Google Forms. The targeted population in this research was individuals within health-care organizations in Jordan. The convenience sample consists of health-care professionals/providers such as general practitioners, who are the current or potential end users of BI systems for health-care purposes. The online survey was distributed to a health-care-related online community to invite the targeted sample to participate. In addition, the objective of the study and questionnaire link was circulated to respondents via their official e-mails and shared on social media channels like WhatsApp groups and the social media pages of health-care organizations. The study population consisted of individuals in various roles within health-care organizations in Jordan, including resident physicians, attending physicians, chief physicians, radiologists and others in the health-care industry. The first page of the online survey included a brief overview of BI based on earlier literature and a 1-min-long video explaining how BI is used in the health-care industry. The participants were requested to rate themselves regarding awareness and understanding of the BI technology, they will continue the following sections only after they confirm that they understand what is BI systems and how such systems could be used in health-care scenarios. Data collection lasted for two months (July–August 2023). All surveys that had the same response to every question or had an excessive number of missing values were disregarded after careful examination. A convenience non-probability sampling method was used as the study’s population (sample frame) was not accessible. Convenience sampling describes “the collection of information from members of the population who are conveniently available to provide it” (Sekaran and Bougie, 2016). Of the 514 returned questionnaires, 434 were valid for data analysis, yielding an 84.43% completion rate. The research approach used in the current study is “structural equation modeling”; hence, a sample of 434 respondents was considered adequate, Table 1 depicts the demographic profiles of the study sample. This sample size is consistent with Kline’s (2015) recommendations, according to which a sample of at least 200 individuals, or five times the number of parameters in the research instrument (questionnaire), is required to be used for this type of study (Kline, 2015).

The survey’s initial version of the questions was written in English, and then, it was translated into Arabic because Jordanian health-care professionals speak Arabic as their mother tongue. The translated version was approved by BI experts, IS academic lecturers, and linguist faculty members who work in Jordanian universities. This study applied a “back translation” practice due to the fact that it is one of the commonly used techniques in international studies (Saunders, Thornhill, and Lewis, 2019). This method provides more quality assurance checks and ensures that every original piece is appropriately captured during the translation process (Ozolins, 2008). A pilot study was conducted to evaluate if the

			Business intelligence
Demographic categories	Frequency	%	
<i>Gender</i>			
• Male	223	51.3	
• Female	211	48.7	
<i>Age</i>			
• 18-29	78	17.97	2147
• 30-39	221	50.92	
• 40-49	129	29.71	
• +50	6	1.4	
<i>Type of health-care organization</i>			
• Public	243	55	
• Private	191	45	
<i>Educational level</i>			
• Undergraduate	363	83.6	
• Master	68	15.9	
• PhD	3	0.5	
<i>Medical professional title</i>			
• Resident physician	146	33.64	
• Attending Physician	107	24.65	
• Chief physician	95	21.9	
• Radiologist	82	18.8	
• others	4	0.92	

Table 1.
Demographic profiles
of the targeted
participants

Source: Created by author

survey items were relevant, simple, and understandable for the study participants. For the pretest, 36 Jordanian medical professionals and BI specialists were selected, and they provided suggestions on the survey questions that were used to modify the measurement instrument and develop the final version. A five-level Likert scale was used for all measurements, with “strongly disagree” (1) being the lowest and “strongly agree” (5) being the highest.

4.2 Measurement

To ensure that the items in the study instrument are valid and reliable, all the scales for this study were adapted from existing UTAUT-related literature as well as earlier empirical studies on “rational decision-making culture,” “perceived threat to professional autonomy” and “medico-legal risk” (see Appendix). Five items were used to assess PE, whereas four items were used to assess EE, SI and FC and BI was measured based on three items. The questionnaire items of these constructs were adapted from (Venkatesh *et al.*, 2003, 2012a, 2012b) and related literature (Fan *et al.*, 2020; Jaklič *et al.*, 2018; Prakash and Das, 2021; Zha *et al.*, 2022). Additionally, three

items were used to evaluate RDC and these were adapted from (Puklavec *et al.*, 2018; Youssef *et al.*, 2022). PTA was measured based on six items adapted from the work of (Walter and Lopez, 2008). Finally, four items were adapted from Chung and Zink (2017) and Prakash and Das (2021) were used to measure MLR. A five-point Likert scale, with the lowest value being “strongly disagree” (1) and the largest being “strongly agree” (5), was used to assess the items that linked to each variable in the suggested research framework. Gender, age, type of healthcare organization, educational level and medical professional title are the demographic information collected about the study participants based on a nominal scale.

5. Data analysis and results

5.1 Demographic profiles

As shown in Table 1, among the participants, 51.3% were males and 48.7% were females. According to the demographic profile, participants are relatively young, with 68.89% being under 39. Regarding their educational level, 83.6% of the study sample held an undergraduate degree, 15.9% of them held a master’s degree, and only 0.5% of them held a PhD degree in their profession. Of the participants, 55% were working in a public healthcare organization, while the largest category in terms of “medical professional title” was “Resident physician” with 33.64%.

5.2 Testing normal distribution

As stated by Hair *et al.* (2014), it is crucial to assess the normality of data in multivariate analysis. Put simply, if the data does not follow a normal distribution, it can potentially impact the accuracy and consistency of the findings. In this study, we used the Jarque–Bera test, which examines the skewness and kurtosis of the data to see if it follows a normal distribution. The skewness value quantifies the degree of symmetry in the distribution (Pallant, 2010). A negative skewness suggests a rightward movement in the distribution, while a positive skewness indicates a leftward shift. Kurtosis is a statistical measure that indicates the degree of peakedness or flatness of a distribution (Pallant, 2010). A positive kurtosis number signifies a distribution that is more peaked, whereas a negative value indicates a distribution that is flatter. The acceptable range for the skewness-kurtosis value is between −2.58 and +2.58. Based on this suggestion, it was determined that all the data points in the data set for the study sample followed a normal distribution, with values falling within the range of ± 2.58. More specifically, the skewness and kurtosis indices have a range of −0.578 to −0.821 and −0.303 to 0.761, respectively.

Table 2.
Descriptive statistics
of the suggested
variables

Constructs	Mean	SD	Cronbach's alpha (α)
PE	3.288	0.936	0.893
Effort expectancy (EE)	3.324	0.921	0.942
SI	2.974	0.817	0.827
FC	3.387	0.866	0.951
BI	3.099	1.049	0.817
RDC	3.47	1.01	0.89
PTA	3.251	0.901	0.922
MLR	3.852	0.849	0.858

Source: Created by author

5.3 Descriptive statistics

The descriptive analysis for each variable in the developed research framework is presented in Table 2. All mean values were higher than 2.947 which shows that the majority of the study sample provided positive replies to the variables that are investigated in this research. Also, Cronbach's α statistics revealed that all factors revealed strong internal reliability.

5.4 Measurement model

To examine the link between the variables in the suggested framework, this study used a two-step methodology. The authors verified the proposed hypothesized relationships in the "structural model" after first analyzing the "measurement model" to determine the validity and reliability of the study instrument (Hair *et al.*, 2014). The hypothesized associations between the factors in the proposed study framework were examined based on "confirmatory factor analysis" using AMOS 25.0 (Byrne, 2016). Maximum likelihood estimation was performed to estimate the model's parameters, and variance-covariance matrices were used in all examinations (Hair *et al.*, 2014). At this point, it is suggested to use some fit indices to assess the postulated model's goodness-of-fit (Hair *et al.*, 2014; Kline, 2015). These indices include χ^2/df , besides at least one absolute index, such as the "root mean square error of approximation (RMSEA)," "standardized root mean square residual (SRMR)," "goodness-of-fit index (GFI)" and "adjusted goodness-of-fit index (AGFI)" and one incremental index, such as "normed fit index (NFI)" and "comparative fit index (CFI)" (Hair *et al.*, 2014; Kenny, 2020; Kline, 2015). The fit indices and acceptable levels of each are shown in Table 3; these illustrate that the model fit for both "measurement and structural models" is sufficient.

Hair *et al.* (2014) stated that reliability, "convergent validity" and "discriminant validity" are measured based on "composite reliability" (CR) and "average variance extracted" (AVE). Hair *et al.* (2014) stated that acceptable "discriminant validity" is indicated by the total AVE of the average value of the constructs being higher than their correlation values, while adequate reliability is indicated by CR being above 0.7, AVE being above 0.5 and CR being greater than AVE. The variables revealed sufficient reliability and convergent validity, as demonstrated by the results in Table 4, which showed that the AVE of all the constructs was higher than 0.612 and higher than 0.848 for CR. Moreover, all measured variables demonstrated sufficient discriminant validity because the square root of AVE is greater than their correlation values.

Fit indices	Measurement model	Structural model	Acceptable value
χ^2/df	2.445	2.449	≤ 3.00
GFI	0.909	0.911	≥ 0.9000
AGFI	0.864	0.865	≥ 0.8000
CFI	0.928	0.927	≥ 0.9000
SRMR	0.042	0.043	≤ 0.05
NFI	0.923	0.919	≥ 0.9000
RMSEA	0.062	0.062	≤ 0.080

Source: Created by author

Table 3.
Model fit indices

Table 4.
“Construct
reliabilities,
convergent validity
and discriminant
validity”

5.5 Structural model
The next step is evaluating the structural model to validate the hypothesized relationships. The results in [Table 5](#) indicate that professionals’ intentions to use BI for health-care purposes were significantly affected by PE, SI, FC, MLR, RDC and PTA. Also, an insignificant effect of PTA on PE was revealed based on the results of statistical analysis. These constructs explained 68% of the variance (R^2) in the individuals’ intentions to use BI-based health-care systems.

6. Discussion

This study investigated the determinants of health-care professionals’ intentions to use BI systems in health-care practices. This section discusses the statistical results as well as the inferences drawn from this investigation.
Professionals with a positive inclination to use BI technology for health-care scenarios are more likely to believe that BI systems could assist them in making diagnoses and other health-care-related decisions more quickly and accurately, boost productivity and reduce workload (*H1* supported). Furthermore, the findings indicate that medical practitioners are more likely to have positive intentions to use BI systems if they feel that their peers, colleagues, professional bodies (medical organizations and groups) and important others support the adoption of such technology in medical practices (*H3* Supported). The results also proved that there is an association between FC and behavioral intentions toward using

	CR	AVE	PE	EE	SI	FC	BI	RDC	PTA	MLR
<i>PE</i>	0.920	0.682	<i>0.825</i>							
<i>EE</i>	0.889	0.715	0.519	<i>0.845</i>						
<i>SI</i>	0.881	0.686	0.672	0.483	<i>0.828</i>					
<i>FC</i>	0.958	0.850	0.823	0.749	0.833	<i>0.921</i>				
<i>BI</i>	0.926	0.795	0.751	0.565	0.658	0.847	<i>0.891</i>			
<i>RDC</i>	0.931	0.621	0.376	0.359	0.418	0.538	0.378	<i>0.788</i>		
<i>PTA</i>	0.898	0.661	−0.149	−0.133	−0.320	−0.331	−0.439	−0.439	<i>0.813</i>	
<i>MLR</i>	0.848	0.612	−0.323	−0.384	−0.344	−0.252	−0.468	−0.224	0.118	<i>0.782</i>

Notes: **“factor correlation matrix with the square root of AVE on the diagonal line (shown as italic)”
Sources: Created by author

Table 5.
Structural model
(hypotheses testing)

Hypotheses	Path	Path coefficient (β)	Results
<i>H1</i>	PE >> BI	0.179**	Supported
<i>H2</i>	EE >> BI	0.003	Not supported
<i>H3</i>	SI >> BI	0.155**	Supported
<i>H4</i>	FC >> BI	0.234**	Supported
<i>H5</i>	MLR >> INT	−0.145	Supported
<i>H6</i>	RDC >> INT	0.451***	Supported
<i>H7</i>	PTA >> IN	−0.281	Supported
<i>H8</i>	PTA >> PE	−0.009	Not supported

Source: Created by author

BI systems (*H4* supported). FC is acknowledged as a highly valued prerequisite for all study participants. These organizational variables are significant for health-care professionals; for example, whether the IT infrastructure is adequate to support the adoption of BI or whether the staff receives appropriate training. The study's conclusions support the findings of earlier investigations, including (Hossain *et al.*, 2019; Moudud-Ul-Huq *et al.*, 2021; Zha *et al.*, 2022; Alkhwaldi, 2023).

Interestingly, the hypothesis (*H2*) has been rejected as EE has unexpectedly been revealed to be an insignificant predictor of BI usage intentions. While this result is inconsistent with the proposition of the original UTAUT (Venkatesh *et al.*, 2003), it corroborates the findings of Prakash and Das (2021) and Kašparová *et al.* (2023) that the IT “ease of use” is an insignificant predictor of usage behavior. A recent research conducted by Zha *et al.* (2022) also reported similar results in the field of accepting “clinical decision support systems” (CDSS) by nurses. This could be because medical practitioners in general are accustomed to working with extremely complicated systems in their daily medical and health-care procedures, and the system's simplicity of use may not be a significant consideration when making adoption-related decisions. Another explanation could be that medical practitioners thought BI technology would be as simple to use as their existing systems and did not anticipate any difficulties.

This research offers an empirical result that RDC has a significant and positive impact on the individual's behavioral intention toward BI in the context of health care (*H5* supported). This finding was consistent with previous literature, for instance (Popovič *et al.*, 2012; Puklavec *et al.*, 2018; Youssef *et al.*, 2022). According to the findings of this study, the adoption of BI technologies in the health-care industry is strongly influenced by a robust “rational decision-making culture.” Organizations that place a high value on a culture of information-driven decision-making are more inclined to adopt BI tools and use them wisely to improve performance and make decisions. It encourages information to be shared and distributed throughout departments, units and individuals inside organizations. Because it makes it effortless to integrate BI into current workflows and processes and promotes the usage of data-driven insights for effective decision-making, this collaborative and knowledge-sharing environment is ideal for the adoption of BI.

The findings validated the relationship suggested in this study, that medical practitioners' PTA plays a significant role in their intention to use BI systems in health-care organizations and confirms the conclusions made by the earlier literature (Sambasivan *et al.*, 2012; Walter and Lopez, 2008). The findings illustrate that if the practitioner perceives BI systems as a key threat to their “professional autonomy” then their intentions to use BI become noticeably low and accordingly the medical practitioners may decide not to use BI systems once it is adopted by their organizations. The relationship between the PTA and PE is revealed to be insignificant. The rational explanation for this result can be attributed to that a small proportion of the respondents (i.e. medical practitioners) have actually used BI systems for health-care decisions.

The statistical analysis also shows that there was a negative impact of PTA on intentions to use BI systems. Furthermore, previous studies on traditional health-care applications confirmed that medical practitioners had higher intentions to adopt such applications. This could possibly attributed to the basic differences in the functionalities between the traditional health-care systems and BI health-care systems. The key function of traditional health-care systems is to increase the health-care organization's productivity. It does not (or in a very limited manner) organize the health-care knowledge. Such systems might free medical professionals from the busy day of data collecting and consolidating, and

in turn, permit them to generate knowledge and to be more involved in providing advice to patients. This impact is indeed revealed in a number of studies investigating health-care systems. In contrast, the key function of BI-assisted systems is to enhance the medical decision-making process by developing rules, techniques and recommendations that offer guidelines for medical professionals' behaviors. This method establishes knowledge that was previously only known by medical professionals and is important for health-care decision-making. IT/IS (e.g. BI-based systems) that codify more knowledge may be seen as more dangerous to "professional autonomy" as the potential result of knowledge codification is outside access to a profession's once-exclusive realm, which in turn affects "professional autonomy." Consequently, medical professionals might be more resistant to BI systems in health-care organizations.

This study confirms that medical users' intentions toward using BI systems were driven by "medico-legal" risk. This indicates that BI users within health-care organizations' concerns about ambiguity outweigh BI system performance, and the potential legal repercussions of adopting the unreliable/error-prone advice of BI technology would further impede their engagement with innovative BI applications. Arguments reported by the articles on BI/AI systems for health care ([Chung and Zink, 2017](#); [Mettler and Vimarlund, 2009](#); [Schweikart, 2020](#)) are also in line with this result.

7. Theoretical and practical implications

7.1 Theoretical implications

Third, the current research contributes to the studies on IT/IS adoption from users' perspective, specifically, the stream of literature that investigated under-researched topics such as the use of BI systems in health-care settings. Despite the significance of this state-of-art technology, very little is known about the antecedents of users' intentions toward this technology particularly in developing countries (e.g. [Jaradat et al., 2022](#)). This research extended the UTAUT in the setting of BI systems by revealing the additional contextual variables affecting the BI usage behavior in the health-care area, namely, "medico-legal" risk, "perceived threat to professional autonomy" and "rational decision-making culture."

From a theoretical perspective and theory building, this study contributes to IT/IS acceptance theories, elucidating user's intentions to use innovative technologies. This study has made an effort to determine vital factors from the individuals' acceptance studies by applying the prevalent UTAUT as a base theoretical framework. As the UTAUT is a general model and could not consider the unique features of health-care professionals, this study framework has been enhanced to better explain professionals' BI usage behavior in a health-care context. The framework proposed in this study could explain 68% of the variance in individuals' intention to use BI-based health-care systems, which is higher than the variance explained in "behavioural intentions" (56%) using the original version of UTAUT ([Venkatesh et al., 2003](#)).

Also, this study contributes by operationalizing and examining an instrument for measuring the new variable "medico-legal risk" in the field. Even though these articles ([Chung and Zink, 2017](#); [Mettler and Vimarlund, 2009](#); [Schweikart, 2020](#)) have described "medico-legal" liability as a significant concern associated with the adoption of AI/BI-based systems for medical and health-care decisions, to the best of the author's knowledge, this research is the first to offer empirical evidence regarding the effect of this inhibiting construct on BI adoption for health-care decision-making.

7.2 Practical implications

The results of this research have significant technology design implications: "Performance expectancy" is not merely a matter of functionality. For users in the health sector, sophisticated systems will not be used if the systems do not follow the natural flow of the users' daily jobs, or

if it could be perceived by the users as threatening their “professional autonomy.” The decision of health-care organizations to accept and use a novel IT/IS (e.g. BI systems) needs to be made with consideration to health-care professionals’ priorities regarding their jobs’ nature.

Likewise, a strong correlation between SI and “behavioural intention” suggests that marketers and the providers of BI technology should invest more in building connections with hospitals, senior leadership in the health-care industry, national and local medical professional associations, and other organizations to guarantee a quicker adoption of BI-based health-care systems. Adoption will be boosted, for instance, by setting up practical training courses for practitioners on BI systems and facilitating in-person meetings with medical professionals engaged in BI systems research and development in association with professional groups. Additionally, more resources ought to be allocated to relationship-building initiatives with professional health-care associations and important others to improve the acceptance and adoption of BI technologies.

FC has also a significant influence on BI acceptance in the health-care context. Thus, revealing a positive image to users, thinking more about users’ requirements and desires and developing their usage value are needed for vendors and developers of BI systems. Thus, vendors and developers of BI systems have to establish a positive image, consider more about the demands and wants of health-care professionals and increase their usage value. It is recommended for vendors of BI systems in collaboration with IT staff in health-care organizations to provide instruction on using BI, manuals and trials (either in physical or digital form). These health-care organizations must have the financial resources to purchase the necessary software and equipment to facilitate the use of BI systems. To use these cutting-edge systems, health-care professionals also need to possess a certain set of IT skills and knowledge.

Before implementing BI, health-care organizations should evaluate their preparedness and capacity to administer and execute BI systems. This involves assessing the staff’s abilities, the organization’s data management procedures and its technology infrastructure. Health-care organizations need to invest in training initiatives to improve staff members’ technical proficiency and foster a “culture of data-driven decision-making.” Furthermore, the adoption of BI is significantly influenced by the “rational decision-making culture.” Health-care providers must establish a workplace that rewards and encourages this particular type of decision-making culture. Campaigns for internal communication, training initiatives and the establishment of clear measurement metrics and objectives based on data-driven insights can all help achieve this.

“Professional autonomy” is an important trait of health-care practitioners. The quality of health care offered to a patient could be significantly developed by adopting and using BI in the health-care industry. With a comprehensive understanding of the key variables that affect medical professionals’ intentions to adopt BI systems in a developing nation, such as PTA, the managers and authorities need to: offer a work setting that promotes knowledge-sharing behavior between professionals and their subordinates; involve professionals in making a decision about adopting BI systems; provide proper training to the professionals; and choose a BI system that offers the opportunity for interaction between the medical professionals and the system while making medical decisions and giving advice to patients. These steps could help to promote professional autonomy for medical professionals. Thus, this could improve the acceptance of BI systems by medical professionals which could have a positive effect on the quality of health care in general.

Finally, the respondents’ perceptions of “medico-legal” risk are revealed to contribute negatively and significantly to users’ intentions to use BI in health care. Considering that current legislation is insufficient to address medical misconduct related to BI use, and malpractice, authorities, policymakers and governments are required to develop new, adequate, predictable legal standards and frameworks for BI-associated medical

malpractice. By doing this, the usage of BI-based decisions will be promoted in medical diagnosis and other decisions in the health-care industry. This also leads to lower levels of perceived risk among medical professionals.

8. Limitations and future research directions

Finally, there are limitations to this research. First, self-reported investigations were the main source of the data used in this research, which may be biased and inaccurate. The validity of the results could be impacted by participant replies that were influenced by memory recall problems or social desirability bias. To improve the reliability and generalizability of the findings, future studies could use several sources of data (data triangulation) or objective measures. Second, this paper focused on investigating variables influencing BI acceptance and use in the health-care sector. It would be worthwhile to replicate this study in various sectors to derive more perceptive and profound insights. Third, as the respondents in this research came from Jordan, the findings may not be generalizable to other nations and cultures. Future research may use the suggested conceptual model in a variety of cultural settings, as well as compare and contrast different nations and cultures, to validate the proposed conceptual model in different societies. Fourth, the cross-sectional design of this research makes it more difficult to prove causality and ascertain the temporal order of events. Experiments or longitudinal studies may offer more reliable insights into the dynamics of BI acceptance and use over time and the causal associations between factors. Finally, further investigations should be carried out to complement the conceptual framework proposed in this research by considering other pertinent constructs and theories. It should be noted that additional theories, such as SCT and perceived value theory, as well as additional factors related to the setting, may also play a role in users' deliberate adoption and should be further explored in future studies. In Arab countries, cultural characteristics, such as cultural context considerations, can significantly influence the success of adopting BI, particularly in the health-care sector. Hofstede (2017) suggests that masculinity, collectivism, power distance and uncertainty avoidance can strongly influence the pace at which IT/IS is accepted and adopted.

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Factor		
Performance expectancy	PE1	Using BI systems would enable me to accelerate my diagnosis and health-care-related decisions
	PE2	Using BI systems would enhance my health-care work performance
	PE3	Using BI systems would promote my diagnosis and other health-care-related decisions level
	PE4	Using BI systems would release my health-care work pressure
	PE5	Overall, I find BI systems useful for health-care purposes
Effort expectancy	EE1	I believe learning to use BI-based health-care systems would be easy for me
	EE2	It would be easy for me to become skillful at using the BI-based health-care systems
	EE3	I think the diagnosis and health-care decision-making process of BI systems is clear and understandable
	EE4	Overall, I believe that the BI-based health-care systems would be easy to use
Social influence	SI1	People who influence my behavior think that I should use BI-based health-care systems
	SI2	People who are important to me think that I should use BI-based health-care systems
	SI3	Senior management in my health-care organization thinks that I should use BI-based health-care systems
	SI4	In general, the organization has promoted the use of BI-based health-care systems
Facilitating conditions	FC1	I can get help from others when I have trouble in using the BI-based health-care systems
	FC2	During my work, BI-based health-care systems work steadily
	FC3	My organization provides adequate training on the use of BI-based health-care systems
	FC4	I have the resources necessary to use BI-based health-care systems
Behavioral intention	BI1	I intend to use BI-based health-care systems in the future
	BI2	I would like to continue to learn more about BI-based health-care systems
	BI3	I would like to recommend BI-based health-care systems to others
Rational decision-making culture	RDC1	My organization encourages to make informed decisions
	RDC2	My organization encourages looking for data/information to inform decision-making
	RDC3	My organization shows organization-wide respect for measuring and evaluating evidence when making decisions

(continued)

Table A1.
Questionnaire

Table A1.

Factor		
Perceived threat to professional autonomy	PTA1	Using BI systems may decrease my control over health-care decisions (Walter and Lopez, 2008)
	PTA2	Using BI systems may decrease my professional discretion over patients' care decisions
	PTA3	Using BI systems can decrease my control over each step of the patient care process
	PTA4	Using BI systems may increase the monitoring of my diagnostic and therapeutic decisions by non-providers
	PTA5	Using BI systems may decrease my control over the allocation of scarce resources
	PTA6	I would find BI systems advantageous for the medical profession as a whole
Medico-legal risk	MLR1	I am concerned that the present liability laws will hold the doctor responsible for any harm/injury arising from an incorrect diagnosis decision suggested by BI-aided decisions (Chung and Zink, 2017; Prakash and Das, 2021)
	MLR2	I am concerned that the present liability laws do not adequately protect doctors from the potential risks (injury/harm to the patient, litigations) arising from the use of BI-aided decisions
	MLR3	I am worried that in case of injury or harm to the patient (resulting from the use of BI-aided decisions), the present liability laws are not clear on who will bear the damages
	MLR4	I am concerned that the manufacturers try to shield themselves from legal liability of harm/injury (resulting from the use of BI-aided decisions) through product disclaimers*

About the author

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